



حوزه آموزش

# سند مسیر یادگیری و منابع آموزشی Java

تهیه‌کننده: کمیته آموزش معاونت توسعه محصول

تاریخ: ۱۴۰۴/۱۲/۲۷

شماره مستند: ۱/۰/۱

این سند از جمله اسناد **درون سازمانی** داتین است.

تکمیل جداول زیر تنها برای مستندات سیستمی الزامی است.

| شناسنامه سند        |            |
|---------------------|------------|
| حوزه کاربردی مستند  | حوزه آموزش |
| زمان بازنگری سند    |            |
| تاریخ منسوخ شدن سند |            |

| تاییدیه سند        |                                |            |  |
|--------------------|--------------------------------|------------|--|
| نام و نام خانوادگی | سمت سازمانی                    | تاریخ      |  |
| تدوین کننده        | کمیته آموزش معاونت توسعه محصول | ۱۴۰۴/۰۵/۳۱ |  |
| تأییدکننده         |                                |            |  |
| تصویب کننده        |                                |            |  |

| جدول تغییرات سند |              |       |                                                  |              |
|------------------|--------------|-------|--------------------------------------------------|--------------|
| ردیف             | تاریخ ویرایش | نسخه  | شرح ویرایش                                       | ویرایش کننده |
| ۰۱               | ۱۴۰۴/۱۲/۲۷   | ۱/۰/۱ | تغییرات در جدول منابع و انتقال منابع به پاداسیپس | کمیته آموزش  |
|                  |              |       |                                                  |              |
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## فهرست مطالب

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## مقدمه

این سند با هدف استانداردسازی و شفافسازی مسیر رشد و یادگیری برای تمام کارشناسان در سطوح مختلف (کاردان، کارشناس، کارشناس ارشد، عالی) تهیه شده است. هدف ما ایجاد یک نقشه راه مشخص است تا هر فرد بداند برای ارتقا به سطح بعدی، به چه دانش و مهارت‌هایی نیاز دارد.

این مستند از سه بخش اصلی تشکیل شده است:

۱. **فهرست جامع منابع آموزشی:** یک کتابخانه مرکزی از تمام منابع مورد تأیید کمیته آموزش.
۲. **مسیرهای یادگیری تفکیکی:** جداول مجزا برای هر سطح شغلی که مشخص می‌کند چه بخش‌هایی از کدام منابع باید مطالعه شود.
۳. **پیوست (سرفصل‌های تفصیلی):** جزئیات و عناوین فصول هر منبع برای مراجعه دقیق‌تر.

## ۱ بخش اول: فهرست جامع منابع آموزشی

این جدول به عنوان یک کتابخانه مرکزی برای تمام منابع آموزشی عمل می‌کند. شناسه‌های موجود در این جدول در بخش مسیرهای یادگیری استفاده خواهند شد.

| جدول ۱: فهرست جامع منابع آموزشی |                 |             |                                                      |                                                                              |                                                                                                                                                                                                                                   |
|---------------------------------|-----------------|-------------|------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| حوزه                            | شناسه منبع (ID) | نوع منبع    | عنوان منبع                                           | نویسنده/مدرس                                                                 | لینک/محل دسترسی                                                                                                                                                                                                                   |
| Data Structures                 | 01              | کتاب        | Data Structures and Algorithms in Java               | Robert Lafore                                                                | <a href="https://podspace.pod.ir/api/files/K9BBZTJZKD4V7B6Q">https://podspace.pod.ir/api/files/K9BBZTJZKD4V7B6Q</a>                                                                                                               |
|                                 | 02              | کتاب        | Algorithms ,Part I and II                            | Robert Sedgewick and Kevin Wayne                                             | <a href="https://podspace.pod.ir/api/files/1HUOEWPY9DEVQRUY">https://podspace.pod.ir/api/files/1HUOEWPY9DEVQRUY</a>                                                                                                               |
|                                 | 03              | کتاب        | Introduction to Algorithms (CLRS)                    | Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein | <a href="https://podspace.pod.ir/api/files/BBJG9P8YWR5E70XN">https://podspace.pod.ir/api/files/BBJG9P8YWR5E70XN</a>                                                                                                               |
|                                 | 04              | کتاب        | داده ساختارها و الگوریتم‌ها                          | دکتر محمد قدسی                                                               | <a href="https://podspace.pod.ir/api/files/H2F6DOREKCEXV872">https://podspace.pod.ir/api/files/H2F6DOREKCEXV872</a>                                                                                                               |
|                                 | 05              | دوره آنلاین | Data Structures Optimizing Performance               | Coursera                                                                     | <a href="https://www.coursera.org/learn/data-structures-optimizing-performance?specialization=java-object-oriented">https://www.coursera.org/learn/data-structures-optimizing-performance?specialization=java-object-oriented</a> |
|                                 | 06              | دوره آنلاین | Data Structures and Algorithms: Deep Dive Using Java | Udemy                                                                        | <a href="https://www.udemy.com/course/data-structures-and-algorithms-deep-dive-using-java/">https://www.udemy.com/course/data-structures-and-algorithms-deep-dive-using-java/</a>                                                 |
|                                 | 07              | سایر        | Data Structures                                      | -                                                                            | <a href="https://www.geeksforgeeks.org/data-structures/">https://www.geeksforgeeks.org/data-structures/</a>                                                                                                                       |
|                                 | 08              | دوره آنلاین | Data Structures & Algorithms in Java                 | Mr. Vishwa Mohan                                                             | <a href="https://www.mygreatlearning.com/academy/learn-for-free/courses/data-structures-and-algorithms-in-java">https://www.mygreatlearning.com/academy/learn-for-free/courses/data-structures-and-algorithms-in-java</a>         |
| Algorithms                      | 03              | کتاب        | Introduction to Algorithms (CLRS)                    | Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein | <a href="https://podspace.pod.ir/api/files/BBJG9P8YWR5E70XN">https://podspace.pod.ir/api/files/BBJG9P8YWR5E70XN</a>                                                                                                               |
|                                 | 01              | کتاب        | Data Structures and Algorithms in Java               | Robert Lafore                                                                | <a href="https://podspace.pod.ir/api/files/K9BBZTJZKD4V7B6Q">https://podspace.pod.ir/api/files/K9BBZTJZKD4V7B6Q</a>                                                                                                               |
|                                 | 02              | کتاب        | Algorithms ,Part I and II                            | Robert Sedgewick and Kevin Wayne                                             | <a href="https://podspace.pod.ir/api/files/1HUOEWPY9DEVQRUY">https://podspace.pod.ir/api/files/1HUOEWPY9DEVQRUY</a>                                                                                                               |
|                                 | 09              | دوره آنلاین | Codio Java Algorithms                                | Coresera                                                                     | <a href="https://www.coursera.org/learn/codio-java-algorithms">https://www.coursera.org/learn/codio-java-algorithms</a>                                                                                                           |
|                                 | 10              | دوره آنلاین | Java Algorithms                                      | Code Academy                                                                 | <a href="https://www.codecademy.com/learn/java-algorithms">https://www.codecademy.com/learn/java-algorithms</a>                                                                                                                   |
|                                 | 11              | سایر        | Fundamental of Algorithms                            | Geeks for Geeks                                                              | <a href="https://www.geeksforgeeks.org/fundamentals-of-algorithms/">https://www.geeksforgeeks.org/fundamentals-of-algorithms/</a>                                                                                                 |
|                                 |                 |             |                                                      |                                                                              |                                                                                                                                                                                                                                   |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------|-------------|----|-----------------|
| <a href="https://www.mygreatlearning.com/academy/learn-for-free/courses/data-structures-and-algorithms-in-java">https://www.mygreatlearning.com/academy/learn-for-free/courses/data-structures-and-algorithms-in-java</a> | Mr. Vishwa Mohan                                                                   | Data Structures & Algorithms in Java                   | دوره آنلاین | 08 |                 |
| <a href="https://podspace.pod.ir/api/files/H2GK41UNFEAI4HU6">https://podspace.pod.ir/api/files/H2GK41UNFEAI4HU6</a>                                                                                                       | Brett McLaughlin, Gary Pollice, and David West                                     | Head First Object-Oriented Analysis and Design         | کتاب        | 12 | Object Oriented |
| <a href="https://podspace.pod.ir/api/files/LM0VG10006XVPLLT">https://podspace.pod.ir/api/files/LM0VG10006XVPLLT</a>                                                                                                       | Bloch, Joshua                                                                      | Effective Java                                         | کتاب        | 13 |                 |
| <a href="https://podspace.pod.ir/api/files/CC5BMYLKPTTFM97C">https://podspace.pod.ir/api/files/CC5BMYLKPTTFM97C</a>                                                                                                       | Robert C. Martin                                                                   | Clean Code: A Handbook of Agile Software Craftsmanship | کتاب        | 14 |                 |
| <a href="https://www.udemy.com/course/oop-learnit/">https://www.udemy.com/course/oop-learnit/</a>                                                                                                                         | Udemy                                                                              | OOP Learn It                                           | دوره آنلاین | 15 |                 |
| <a href="https://www.udemy.com/course/ap-computer-science-b-java-object-oriented-programming/">https://www.udemy.com/course/ap-computer-science-b-java-object-oriented-programming/</a>                                   | Udemy                                                                              | Java Object Oriented Programming                       | دوره آنلاین | 16 |                 |
| <a href="https://podspace.pod.ir/api/files/H91QKL2W9ST49K52">https://podspace.pod.ir/api/files/H91QKL2W9ST49K52</a>                                                                                                       | Herbert Schildt                                                                    | Java: A Beginner's Guide                               | کتاب        | 17 | Java Core       |
| <a href="https://podspace.pod.ir/api/files/LJJ8GN58U6T3VCPL">https://podspace.pod.ir/api/files/LJJ8GN58U6T3VCPL</a>                                                                                                       | Kathy Sierra and Bert Bates                                                        | Head First Java                                        | کتاب        | 18 |                 |
| <a href="https://podspace.pod.ir/api/files/LM0VG10006XVPLLT">https://podspace.pod.ir/api/files/LM0VG10006XVPLLT</a>                                                                                                       | Bloch, Joshua                                                                      | Effective Java                                         | کتاب        | 13 |                 |
| <a href="https://podspace.pod.ir/api/files/GT6X4D4YT1HEA9ZB">https://podspace.pod.ir/api/files/GT6X4D4YT1HEA9ZB</a>                                                                                                       | Oracle                                                                             | Core Java, Volume I: Fundamentals (Oracle Press Java)  | کتاب        | 19 |                 |
| <a href="https://www.w3schools.com/java/default.asp">https://www.w3schools.com/java/default.asp</a>                                                                                                                       | W3School                                                                           | Java                                                   | دوره آنلاین | 20 |                 |
| <a href="https://www.geeksforgeeks.org/java/">https://www.geeksforgeeks.org/java/</a>                                                                                                                                     | GeeksForGeeks                                                                      | Java                                                   | دوره آنلاین | 21 |                 |
| <a href="https://www.codecademy.com/learn/learn-java">https://www.codecademy.com/learn/learn-java</a>                                                                                                                     | Code Academy                                                                       | Learn Java                                             | دوره آنلاین | 22 |                 |
| <a href="https://techvidvan.com/tutorials/learn-java/">https://techvidvan.com/tutorials/learn-java/</a>                                                                                                                   | TechVidVan                                                                         | Learn Java                                             | دوره آنلاین | 23 |                 |
| <a href="https://www.freecodecamp.org/news/learn-java-free-java-courses-for-beginners/">https://www.freecodecamp.org/news/learn-java-free-java-courses-for-beginners/</a>                                                 | FreeCodeCamp                                                                       | Java for Beginners                                     | دوره آنلاین | 24 |                 |
| <a href="https://www.udacity.com/course/java-programming-basics--ud282">https://www.udacity.com/course/java-programming-basics--ud282</a>                                                                                 | Udacity                                                                            | Java Programming Basics                                | دوره آنلاین | 25 |                 |
| <a href="https://podspace.pod.ir/api/files/5H3L9A2RFA2T4DCC">https://podspace.pod.ir/api/files/5H3L9A2RFA2T4DCC</a>                                                                                                       | Herbert Schildt                                                                    | Java :The Complete Reference                           | کتاب        | 26 | Java Advanced   |
| <a href="https://podspace.pod.ir/api/files/AAVWUSPFMSCZ1PA6">https://podspace.pod.ir/api/files/AAVWUSPFMSCZ1PA6</a>                                                                                                       | Scott Oaks                                                                         | Java Performance: The Definitive Guide                 | کتاب        | 27 |                 |
| <a href="https://podspace.pod.ir/api/files/1ANC4LBICRR8RV6S">https://podspace.pod.ir/api/files/1ANC4LBICRR8RV6S</a>                                                                                                       | Brian Goetz, Tim Peierls, Joshua Bloch, Joseph Bowbeer, David Holmes, and Doug Lea | Java Concurrency in Practice                           | کتاب        | 28 |                 |
| <a href="https://podspace.pod.ir/api/files/LM0VG10006XVPLLT">https://podspace.pod.ir/api/files/LM0VG10006XVPLLT</a>                                                                                                       | Bloch, Joshua                                                                      | Effective Java                                         | کتاب        | 13 |                 |

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| <a href="https://podspace.pod.ir/api/files/M8S3WBCH82N8SEUW">https://podspace.pod.ir/api/files/M8S3WBCH82N8SEUW</a>                                                               | Alan Mycroft, Mario Fusco, and Raoul-Gabriel Urma                           | Modern Java in Action: Lambdas, Streams, Functional and Reactive Programming  | کتاب        | 29 |                 |
| <a href="https://podspace.pod.ir/api/files/LO8POH6DAN1RSWUD">https://podspace.pod.ir/api/files/LO8POH6DAN1RSWUD</a>                                                               | Oracle                                                                      | Core Java: Advanced Features, Volume 2                                        | کتاب        | 30 |                 |
| <a href="https://www.coursera.org/learn/advanced-java">https://www.coursera.org/learn/advanced-java</a>                                                                           | Coursera                                                                    | Advanced Java                                                                 | دوره آنلاین | 31 |                 |
| <a href="https://www.udemy.com/course/learn-advanced-java">https://www.udemy.com/course/learn-advanced-java</a>                                                                   | Udemy                                                                       | Learn Advanced Java                                                           | دوره آنلاین | 32 |                 |
| <a href="https://www.geeksforgeeks.org/java/">https://www.geeksforgeeks.org/java/</a>                                                                                             | GeeksForGeeks                                                               | Java                                                                          | دوره آنلاین | 21 |                 |
| <a href="https://dzone.com/java">https://dzone.com/java</a>                                                                                                                       | Dzone                                                                       | Java                                                                          | دوره آنلاین | 33 |                 |
| <a href="https://stackify.com/java-tutorials/">https://stackify.com/java-tutorials/</a>                                                                                           | Stackify                                                                    | Java Tutorials                                                                | دوره آنلاین | 34 |                 |
| <a href="https://podspace.pod.ir/api/files/4BGKMXAWMAONPTYQ">https://podspace.pod.ir/api/files/4BGKMXAWMAONPTYQ</a>                                                               | Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides (Gang of Four) | Design Patterns: Elements of Reusable Object-Oriented Software (Gang of Four) | کتاب        | 35 | Design Patterns |
| <a href="https://podspace.pod.ir/api/files/3ZVAP63H7XXPKU8P">https://podspace.pod.ir/api/files/3ZVAP63H7XXPKU8P</a>                                                               | Martin Fowler                                                               | Patterns of Enterprise Application Architecture                               | کتاب        | 36 |                 |
| <a href="https://podspace.pod.ir/api/files/E1IFQLGA6TC426GR">https://podspace.pod.ir/api/files/E1IFQLGA6TC426GR</a>                                                               | Miroslav Wengner                                                            | Practical Design Patterns for Java Developers                                 | کتاب        | 37 |                 |
| <a href="https://refactoring.guru/design-patterns">https://refactoring.guru/design-patterns</a>                                                                                   | Refactoring Guru                                                            | Design Patterns                                                               | دوره آنلاین | 38 |                 |
| <a href="https://www.coursera.org/learn/design-patterns">https://www.coursera.org/learn/design-patterns</a>                                                                       | Coursera                                                                    | Design Patterns                                                               | دوره آنلاین | 39 |                 |
| <a href="https://javadesignpatterns.com/">https://javadesignpatterns.com/</a>                                                                                                     | JavaDesignPatterns                                                          | Design Patterns                                                               | دوره آنلاین | 40 |                 |
| <a href="https://podspace.pod.ir/api/files/BVGHN5Y1I6U5G1M7">https://podspace.pod.ir/api/files/BVGHN5Y1I6U5G1M7</a>                                                               | Christian Bauer, Gavin King, and Gary Gregory                               | Java Persistence with Hibernate                                               | کتاب        | 41 | ORM             |
| <a href="https://podspace.pod.ir/api/files/2U9X56DBSJS2NWWH">https://podspace.pod.ir/api/files/2U9X56DBSJS2NWWH</a>                                                               | Catalin Tudose                                                              | Java Persistence with Spring Data and Hibernate                               | کتاب        | 42 |                 |
| <a href="https://www.udemy.com/course/hibernate-and-jpa-fundamentals/">https://www.udemy.com/course/hibernate-and-jpa-fundamentals/</a>                                           | Udemy                                                                       | Hibernate & JPA Fundamentals                                                  | دوره آنلاین | 43 |                 |
| <a href="https://www.geeksforgeeks.org/hibernate-tutorial/">https://www.geeksforgeeks.org/hibernate-tutorial/</a>                                                                 | GeeksForGeeks                                                               | Hibernate Tutorial                                                            | دوره آنلاین | 44 |                 |
| <a href="https://www.baeldung.com/learn-jpa-hibernate">https://www.baeldung.com/learn-jpa-hibernate</a>                                                                           | Baeldung                                                                    | Learn Jpa Hibernate                                                           | دوره آنلاین | 45 |                 |
| <a href="https://www.digitalocean.com/community/tutorials/hibernate-tutorial-for-beginners">https://www.digitalocean.com/community/tutorials/hibernate-tutorial-for-beginners</a> | Digital Ocean                                                               | Hibernate Tutorial for Beginners                                              | دوره آنلاین | 46 |                 |
| <a href="https://www.udemy.com/course/java-database-connectivity-jdbc-masterclass/">https://www.udemy.com/course/java-database-connectivity-jdbc-masterclass/</a>                 | Udemy                                                                       | Java Database Connectivity JDBC Masterclass                                   | دوره آنلاین | 47 |                 |
| <a href="https://www.vogella.com/tutorials/EclipseLink/article.html">https://www.vogella.com/tutorials/EclipseLink/article.html</a>                                               | Vogella                                                                     | Eclipse Link                                                                  | دوره آنلاین | 48 |                 |

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| <a href="https://www.udemy.com/course/java-persistence-using-mybatis">https://www.udemy.com/course/java-persistence-using-mybatis</a>                                                                           | Udemy                                       | Java Persistence Using Mybatis                                          | دوره آنلاین | 49 |                 |
| <a href="https://www.tutorialspoint.com/mybatis/index.htm">https://www.tutorialspoint.com/mybatis/index.htm</a>                                                                                                 | Udemy                                       | Mybatis                                                                 | دوره آنلاین | 50 |                 |
| <a href="https://www.udemy.com/course/spring-hibernate-tutorial/">https://www.udemy.com/course/spring-hibernate-tutorial/</a>                                                                                   | Udemy                                       | Spring Hibernate Tutorial                                               | دوره آنلاین | 51 |                 |
| <a href="https://podspace.pod.ir/api/files/KK8B4VZX3FSBSKHF">https://podspace.pod.ir/api/files/KK8B4VZX3FSBSKHF</a>                                                                                             | Craig Walls                                 | Spring in Action                                                        | کتاب        | 52 | Spring          |
| <a href="https://podspace.pod.ir/api/files/MLS9AEV7TTXBF6XR">https://podspace.pod.ir/api/files/MLS9AEV7TTXBF6XR</a>                                                                                             | Chris Schaefer, Clarence Ho, and Rob Harrop | Pro Spring 5: An In-Depth Guide to the Spring Framework and Its Tools   | کتاب        | 53 |                 |
| <a href="https://spring.io/">https://spring.io/</a>                                                                                                                                                             | Spring                                      | Spring Documentation                                                    | دوره آنلاین | 54 |                 |
| <a href="https://www.coursera.org/specializations/spring-framework">https://www.coursera.org/specializations/spring-framework</a>                                                                               | Coursera                                    | Spring Framework                                                        | دوره آنلاین | 55 |                 |
| <a href="https://www.udemy.com/course/spring-framework-5-beginner-to-guru/">https://www.udemy.com/course/spring-framework-5-beginner-to-guru/</a>                                                               | Udemy                                       | Spring Framework 5 Beginner to Guru                                     | دوره آنلاین | 56 |                 |
| <a href="https://www.linkedin.com/learning/spring-framework-in-depth">https://www.linkedin.com/learning/spring-framework-in-depth</a>                                                                           | Frank P Moley                               | Spring Framework in Depth                                               | دوره آنلاین | 57 |                 |
| <a href="https://www.digialocean.com/community/tutorials/spring-tutorial-spring-core-tutorial">https://www.digialocean.com/community/tutorials/spring-tutorial-spring-core-tutorial</a>                         | Digital Ocean                               | Spring Core Tutorial                                                    | دوره آنلاین | 58 |                 |
| <a href="https://podspace.pod.ir/api/files/EXHM3IO5T43E7RWP">https://podspace.pod.ir/api/files/EXHM3IO5T43E7RWP</a>                                                                                             | Bill Burke                                  | RESTful Java with JAX-RS 2.0                                            | کتاب        | 59 | Web Service     |
| <a href="https://podspace.pod.ir/api/files/5EG6OB7O32ZKXWOP">https://podspace.pod.ir/api/files/5EG6OB7O32ZKXWOP</a>                                                                                             | Kasun Indrasiri and Danesh Kuruppu          | gRPC: Up and Running                                                    | کتاب        | 60 |                 |
| <a href="https://www.udemy.com/course/java-web-services">https://www.udemy.com/course/java-web-services</a>                                                                                                     | Udemy                                       | Java Web Services                                                       | دوره آنلاین | 61 |                 |
| <a href="https://www.baeldung.com/building-a-restful-web-service-with-spring-and-java-based-configuration">https://www.baeldung.com/building-a-restful-web-service-with-spring-and-java-based-configuration</a> | Baeldung                                    | Building a Restful Web Service with Spring and Java Based Configuration | دوره آنلاین | 62 |                 |
| <a href="https://www.linkedin.com/learning/programming-foundations-apis-and-web-services">https://www.linkedin.com/learning/programming-foundations-apis-and-web-services</a>                                   | Kesha Williams                              | Programming Foundations: APIs and Web Services                          | دوره آنلاین | 63 |                 |
| <a href="https://www.udemy.com/course/learn-graphql-in-java/">https://www.udemy.com/course/learn-graphql-in-java/</a>                                                                                           | Udemy                                       | Learn GraphQL in Java                                                   | دوره آنلاین | 64 |                 |
| <a href="https://www.udemy.com/course/grpc-the-complete-guide-for-java-developers/">https://www.udemy.com/course/grpc-the-complete-guide-for-java-developers/</a>                                               | Udemy                                       | gRPC the Complete Guide for Java Developers                             | دوره آنلاین | 65 |                 |
| <a href="https://grpc.io/docs/languages/java/basics/">https://grpc.io/docs/languages/java/basics/</a>                                                                                                           | gRPC                                        | gRPC Documentation                                                      | دوره آنلاین | 66 |                 |
| <a href="https://podspace.pod.ir/api/files/1OI7OVNLU2HB5CAG">https://podspace.pod.ir/api/files/1OI7OVNLU2HB5CAG</a>                                                                                             | Scott Chacon and Ben Straub                 | Pro Git                                                                 | کتاب        | 67 | Version Control |
| <a href="https://podspace.pod.ir/api/files/A3TA86KMQY5ES332">https://podspace.pod.ir/api/files/A3TA86KMQY5ES332</a>                                                                                             | Jon Loeliger and Matthew McCullough         | Version Control with Git                                                | کتاب        | 68 |                 |
| <a href="https://www.coursera.org/learn/version-control-with-git">https://www.coursera.org/learn/version-control-with-git</a>                                                                                   | Coursera                                    | Version Control with Git                                                | دوره آنلاین | 69 |                 |



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| <a href="https://www.udacity.com/course/introduction-to-version-control--cd0419">https://www.udacity.com/course/introduction-to-version-control--cd0419</a>                                           | Udacity                             | Introduction to Version Control                                                                                                                         | دوره آنلاین | 70 |             |
| <a href="https://www.linkedin.com/learning/git-essential-training-19417064">https://www.linkedin.com/learning/git-essential-training-19417064</a>                                                     | Barbara Forbes                      | Git Essential Training                                                                                                                                  | دوره آنلاین | 71 |             |
| <a href="https://www.codecademy.com/learn/learn-git">https://www.codecademy.com/learn/learn-git</a>                                                                                                   | Code Academy                        | Learn Git                                                                                                                                               | دوره آنلاین | 72 |             |
| <a href="https://podspace.pod.ir/api/files/EHVH29EROCJHD71X">https://podspace.pod.ir/api/files/EHVH29EROCJHD71X</a>                                                                                   | Walter Shields                      | SQL QuickStart Guide: The Simplified Beginner's Guide to Managing, Analyzing, and Manipulating Data With SQL (Coding & Programming - QuickStart Guides) | کتاب        | 73 | Database    |
| <a href="https://podspace.pod.ir/api/files/EDPYSRBBFL1JQGXXH">https://podspace.pod.ir/api/files/EDPYSRBBFL1JQGXXH</a>                                                                                 | Alan Beaulieu                       | Learning SQL: Generate, Manipulate, and Retrieve Data                                                                                                   | کتاب        | 74 |             |
| <a href="https://www.udemy.com/course/the-sql-database-with-java-build-your-first-desktop-app/">https://www.udemy.com/course/the-sql-database-with-java-build-your-first-desktop-app/</a>             | Udemy                               | The SQL Database with Java – Build Your First Desktop App                                                                                               | دوره آنلاین | 75 |             |
| <a href="https://www.coursera.org/specializations/oracle-sql-databases">https://www.coursera.org/specializations/oracle-sql-databases</a>                                                             | Coursera                            | Oracle SQL Databases                                                                                                                                    | دوره آنلاین | 76 |             |
| <a href="https://www.udemy.com/course/the-complete-sql-bootcamp/">https://www.udemy.com/course/the-complete-sql-bootcamp/</a>                                                                         | Udemy                               | The Complete SQL Bootcamp                                                                                                                               | دوره آنلاین | 77 |             |
| <a href="https://www.udemy.com/course/oracle-sql-12c-become-an-sql-developer-with-subtitle/">https://www.udemy.com/course/oracle-sql-12c-become-an-sql-developer-with-subtitle/</a>                   | Udemy                               | Oracle SQL 12C Become an SQL Developer with Subtitle                                                                                                    | دوره آنلاین | 78 |             |
| <a href="https://www.linkedin.com/learning/sql-essential-training-20685933">https://www.linkedin.com/learning/sql-essential-training-20685933</a>                                                     | Walter Shields                      | SQL Essential Training                                                                                                                                  | دوره آنلاین | 79 |             |
| <a href="https://www.geeksforgeeks.org/sql-tutorial/">https://www.geeksforgeeks.org/sql-tutorial/</a>                                                                                                 | GeeksForGeeks                       | SQL Tutorial                                                                                                                                            | دوره آنلاین | 80 |             |
| <a href="https://www.freecodecamp.org/news/learn-sql-free-relational-database-courses-for-beginners/">https://www.freecodecamp.org/news/learn-sql-free-relational-database-courses-for-beginners/</a> | FreeCodeCamp                        | Learn SQL Free Relational Database Courses for Beginners                                                                                                | دوره آنلاین | 81 |             |
| <a href="https://www.w3schools.com/sql/default.asp">https://www.w3schools.com/sql/default.asp</a>                                                                                                     | W3Schools                           | SQL                                                                                                                                                     | دوره آنلاین | 82 |             |
| <a href="https://learnsql.com/">https://learnsql.com/</a>                                                                                                                                             | LearnSQL                            | Learn SQL                                                                                                                                               | دوره آنلاین | 83 |             |
| <a href="https://devgym.oracle.com/pls/apex/dg/class/databases-for-developers-foundations.html">https://devgym.oracle.com/pls/apex/dg/class/databases-for-developers-foundations.html</a>             | DevGym                              | Databases for Developers Foundations                                                                                                                    | دوره آنلاین | 84 |             |
| <a href="https://podspace.pod.ir/api/files/MR1PRLWUL8928HNL">https://podspace.pod.ir/api/files/MR1PRLWUL8928HNL</a>                                                                                   | Benjamin Muschko                    | Gradle in Action                                                                                                                                        | کتاب        | 85 | Build Tools |
| <a href="https://www.oreilly.com/library/view/building-and-testing/9781449306816/">https://www.oreilly.com/library/view/building-and-testing/9781449306816/</a>                                       | Tim Berglund and Matthew McCullough | Building and Testing with Gradle                                                                                                                        | کتاب        | 86 |             |
| <a href="https://udemy.com/course/gradle-for-java-developers/">https://udemy.com/course/gradle-for-java-developers/</a>                                                                               | Udemy                               | Gradle for Java Developers                                                                                                                              | دوره آنلاین | 87 |             |

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| <a href="https://www.udemy.com/course/apache-maven-beginner-to-guru/">https://www.udemy.com/course/apache-maven-beginner-to-guru/</a>                                                                                   | Udemy                                  | Apache Maven Beginner to Guru                          | دوره آنلاین | 88  |                                  |
| <a href="https://www.linkedin.com/learning/learning-gradle">https://www.linkedin.com/learning/learning-gradle</a>                                                                                                       | Benjamin Muschko                       | Learning Gradle                                        | دوره آنلاین | 89  |                                  |
| <a href="https://www.educative.io/courses/build-java-projects-with-maven">https://www.educative.io/courses/build-java-projects-with-maven</a>                                                                           | Educative                              | Build Java Projects with Maven                         | دوره آنلاین | 90  |                                  |
| <a href="https://www.javacodegeeks.com/2018/05/gradle-tutorial.html">https://www.javacodegeeks.com/2018/05/gradle-tutorial.html</a>                                                                                     | JavaCoddeGeeks                         | Gradle Tutorial                                        | دوره آنلاین | 91  |                                  |
| <a href="https://www.geeksforgeeks.org/maven-tutorial/">https://www.geeksforgeeks.org/maven-tutorial/</a>                                                                                                               | GeeksForGeeks                          | Maven Tutrial                                          | دوره آنلاین | 92  |                                  |
| <a href="https://podspace.pod.ir/api/files/31H6TNSRGMM1IEOY">https://podspace.pod.ir/api/files/31H6TNSRGMM1IEOY</a>                                                                                                     | Viktor Farcic and Alex Garcia          | Test-Driven Java Development                           | کتاب        | 93  | Test Automation                  |
| <a href="https://www.simonandschuster.com/books/Contract-Testing-in-Action/Marie-Cruz/In-Action/9781633437241">https://www.simonandschuster.com/books/Contract-Testing-in-Action/Marie-Cruz/In-Action/9781633437241</a> | Marie Cruz (Simon & Schuster)          | Contract Testing in Action                             | دوره آنلاین | 94  |                                  |
| <a href="https://www.udemy.com/course/java-test-automation-engineer-from-zero-to-hero/?couponCode=CMCP SALE24">https://www.udemy.com/course/java-test-automation-engineer-from-zero-to-hero/?couponCode=CMCP SALE24</a> | Udemy                                  | Test Automation Engineer from Zero to Hero             | دوره آنلاین | 95  |                                  |
| <a href="https://www.udemy.com/course/java-and-cucumber-for-automation-testing/">https://www.udemy.com/course/java-and-cucumber-for-automation-testing/</a>                                                             | Udemy                                  | Java and Cucumber for Automation Testing               | دوره آنلاین | 96  |                                  |
| <a href="https://www.udemy.com/course/testing-java-code-with-junit-5-and-mockito/">https://www.udemy.com/course/testing-java-code-with-junit-5-and-mockito/</a>                                                         | Udemy                                  | Testing Java Code With Junit 5 & Mockito               | دوره آنلاین | 97  |                                  |
| <a href="https://www.udemy.com/course/spring-boot-unit-testing/">https://www.udemy.com/course/spring-boot-unit-testing/</a>                                                                                             | Udemy                                  | Spring Boot Unit Testing                               | دوره آنلاین | 98  |                                  |
| <a href="https://www.educative.io/courses/test-automation-framework-">https://www.educative.io/courses/test-automation-framework-</a>                                                                                   | Educative                              | Test Automation Framework                              | دوره آنلاین | 99  |                                  |
| <a href="https://www.coursera.org/specializations/software-testing-automation-selenium-java">https://www.coursera.org/specializations/software-testing-automation-selenium-java</a>                                     | Coursera                               | Software Testing Automation Selenium Java              | دوره آنلاین | 100 |                                  |
| <a href="https://www.coursera.org/projects/api-testing-using-rest-assured-test-automation-tool">https://www.coursera.org/projects/api-testing-using-rest-assured-test-automation-tool</a>                               | Coursera                               | API Testing Using Rest Assured Test Automation Tool    | دوره آنلاین | 101 |                                  |
| <a href="https://www.coursera.org/projects/performance-testing-with-jmeter-for-absolute-beginners">https://www.coursera.org/projects/performance-testing-with-jmeter-for-absolute-beginners</a>                         | Coursera                               | Performance Testing with Jmeter for Absolute Beginners | دوره آنلاین | 102 |                                  |
| <a href="https://www.linkedin.com/learning/postman-essential-training-21969591">https://www.linkedin.com/learning/postman-essential-training-21969591</a>                                                               | Kristin Jackvony                       | Postman Essential Training                             | دوره آنلاین | 103 |                                  |
| <a href="https://www.udemy.com/course/cypress-end-to-end-testing-getting-started/">https://www.udemy.com/course/cypress-end-to-end-testing-getting-started/</a>                                                         | Udemy                                  | Cypress End to End Testing Getting Started             | دوره آنلاین | 104 |                                  |
| <a href="https://www.geeksforgeeks.org/introduction-to-junit-5/">https://www.geeksforgeeks.org/introduction-to-junit-5/</a>                                                                                             | GeeksForGeeks                          | Introduction to Junit 5                                | دوره آنلاین | 105 |                                  |
| <a href="https://podspace.pod.ir/api/files/BJDT80MLIACLM5BQ">https://podspace.pod.ir/api/files/BJDT80MLIACLM5BQ</a>                                                                                                     | Justin Richer and Antonio Sanso        | OAuth 2 in Action                                      | کتاب        | 106 | Authenticat & ion AUTHORIZATI on |
| <a href="https://podspace.pod.ir/api/files/A7TQ7W7FXLRWV7YM">https://podspace.pod.ir/api/files/A7TQ7W7FXLRWV7YM</a>                                                                                                     | Laurentiu Spilca                       | Spring Security in Action                              | کتاب        | 107 |                                  |
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| <a href="https://www.udemy.com/course/java-spring-boot-security-authentication-authorization/">https://www.udemy.com/course/java-spring-boot-security-authentication-authorization/</a>                                                                                             | Udemy                                              | Java Spring Boot Security Authentication Authorizatrion      | دوره آنلاین | 109 |            |
| <a href="https://www.udacity.com/course/identity-access-management-cd0039">https://www.udacity.com/course/identity-access-management-cd0039</a>                                                                                                                                     | Udacity                                            | Identity Access Management                                   | دوره آنلاین | 110 |            |
| <a href="https://www.javatpoint.com/authentication-and-authorization-in-java">https://www.javatpoint.com/authentication-and-authorization-in-java</a>                                                                                                                               | JavaPoint                                          | Authentication & Authorization in Java                       | دوره آنلاین | 111 |            |
| <a href="https://www.geeksforgeeks.org/authentication-and-authorization-in-microservices/">https://www.geeksforgeeks.org/authentication-and-authorization-in-microservices/</a>                                                                                                     | GeeksForGeeks                                      | Authentication & Authorization in Microservices              | دوره آنلاین | 112 |            |
| <a href="https://dzone.com/articles/jwt-authentication-and-authorization-a-detailed-introduction">https://dzone.com/articles/jwt-authentication-and-authorization-a-detailed-introduction</a>                                                                                       | DZone                                              | JWT Authentication & Authorization a Detailed Introduction   | دوره آنلاین | 113 |            |
| <a href="https://www.freecodecamp.org/news/fine-grained-authorization-in-java-and-springboot">https://www.freecodecamp.org/news/fine-grained-authorization-in-java-and-springboot</a>                                                                                               | FreeCodeMap                                        | Fine Grained Authorization in va and Spring boot             | دوره آنلاین | 114 |            |
| <a href="https://www.udemy.com/course/rest-api-security-tls-basicauth-authorization-jwt-oauth-mfa-hmac/">https://www.udemy.com/course/rest-api-security-tls-basicauth-authorization-jwt-oauth-mfa-hmac/</a>                                                                         | Udemy                                              | REST API Security-BasicAuth-Authorization-JWT-OAuth-MFA-HMAC | دوره آنلاین | 115 |            |
| <a href="https://www.udemy.com/course/learn-oauth-2/">https://www.udemy.com/course/learn-oauth-2/</a>                                                                                                                                                                               | Udemy                                              | Learn Oauth2                                                 | دوره آنلاین | 116 |            |
| <a href="https://education.launchcode.org/java-web-dev-curriculum/authentication/reading/intro-to-authentication/index.html">https://education.launchcode.org/java-web-dev-curriculum/authentication/reading/intro-to-authentication/index.html</a>                                 | Launch Code                                        | Intro to Authentication                                      | دوره آنلاین | 117 |            |
| <a href="https://www.educative.io/path/web-authentication-and-authorization">https://www.educative.io/path/web-authentication-and-authorization</a>                                                                                                                                 | Educative                                          | Web Authentication and Authorization                         | دوره آنلاین | 118 |            |
| <a href="https://www.baeldung.com/java-authentication-authorization-service">https://www.baeldung.com/java-authentication-authorization-service</a>                                                                                                                                 | Baeldung                                           | Java Authentication Authorization Service                    | دوره آنلاین | 119 | Monitoring |
| <a href="https://www.oreilly.com/library/view/java-projects/9781789131895/e4725011-b397-4a9f-88f8-8cf1c0fe7a40.xhtml">https://www.oreilly.com/library/view/java-projects/9781789131895/e4725011-b397-4a9f-88f8-8cf1c0fe7a40.xhtml</a>                                               | Sam Brannen                                        | Java Logging                                                 | کتاب        | 120 |            |
| <a href="https://podspace.pod.ir/api/files/GFKCF7ANAU4WVGL7">https://podspace.pod.ir/api/files/GFKCF7ANAU4WVGL7</a>                                                                                                                                                                 | Charity Majors, George Miranda, and Liz Fong-Jones | Observability Engineering: Achieving Production Excellence   | کتاب        | 121 |            |
| <a href="https://www.pluralsight.com/courses/java-se-17-loghttps://alison.com/course/java-logging-frameworks-the-complete-guided-management">https://www.pluralsight.com/courses/java-se-17-loghttps://alison.com/course/java-logging-frameworks-the-complete-guided-management</a> | PluralSight                                        | Java Logging Frameworks the Complete Management Guide        | دوره آنلاین | 122 |            |
| <a href="https://www.baeldung.com/java-logging-intro">https://www.baeldung.com/java-logging-intro</a>                                                                                                                                                                               | Baeldung                                           | Java Logging Intro                                           | دوره آنلاین | 123 |            |
| <a href="https://www.baeldung.com/spring-boot-logging">https://www.baeldung.com/spring-boot-logging</a>                                                                                                                                                                             | Baeldung                                           | Spring boot Logging                                          | دوره آنلاین | 124 |            |

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| <a href="https://www.geeksforgeeks.org/logging-in-java/">https://www.geeksforgeeks.org/logging-in-java/</a>                                                                                       | GeeksForGeeks                            | Logging in Java                                                          | دوره آنلاین | 125 |                              |
| <a href="https://podspace.pod.ir/api/files/ED8P8IFRQEH1RVML">https://podspace.pod.ir/api/files/ED8P8IFRQEH1RVML</a>                                                                               | Radu Gheorghe                            | Elasticsearch in Action                                                  | کتاب        | 126 |                              |
| <a href="https://www.udemy.com/course/complete-elasticsearch-masterclass-with-kibana-and-logstash/">https://www.udemy.com/course/complete-elasticsearch-masterclass-with-kibana-and-logstash/</a> | Udemy                                    | Complete Elasticsearch Masterclass with Logstash and Kibana              | دوره آنلاین | 127 |                              |
| <a href="https://www.oreilly.com/library/view/mastering-prometheus/9781805125662/">https://www.oreilly.com/library/view/mastering-prometheus/9781805125662/</a>                                   | William Hegedus                          | Mastering Prometheus                                                     | کتاب        | 128 |                              |
| <a href="https://prometheus.io/docs/prometheus/latest/getting_started/">https://prometheus.io/docs/prometheus/latest/getting_started/</a>                                                         | Prometheus                               | Prometheus Guide                                                         | دوره آنلاین | 129 |                              |
| <a href="https://www.udemy.com/course/prometheus-course/">https://www.udemy.com/course/prometheus-course/</a>                                                                                     | Udemy                                    | Prometheus Course                                                        | دوره آنلاین | 130 |                              |
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| <a href="https://dzone.com/refcardz/java-caching">https://dzone.com/refcardz/java-caching</a>                                                                                                     | DZone                                    | Java Caching                                                             | دوره آنلاین | 132 |                              |
| <a href="https://www.udemy.com/course/redis-the-complete-developers-guide-p/">https://www.udemy.com/course/redis-the-complete-developers-guide-p/</a>                                             | Udemy                                    | Redis The Complete Developer Guide                                       | دوره آنلاین | 133 |                              |
| <a href="https://dzone.com/refcardz/java-caching">https://dzone.com/refcardz/java-caching</a>                                                                                                     | DZone                                    | Java Caching                                                             | دوره آنلاین | 134 |                              |
| <a href="https://www.baeldung.com/caching-series">https://www.baeldung.com/caching-series</a>                                                                                                     | Baeldung                                 | Caching Series                                                           | دوره آنلاین | 135 |                              |
| <a href="https://www.udemy.com/course/learn-redis-and-utilize-jedis-with-spring-data-redis/">https://www.udemy.com/course/learn-redis-and-utilize-jedis-with-spring-data-redis/</a>               | Udemy                                    | Learn Redis and Utilize Jedis with Spring Data Redis                     | دوره آنلاین | 136 |                              |
| <a href="https://stage-university.redis.com/">https://stage-university.redis.com/</a>                                                                                                             | Stage University                         | Redis                                                                    | دوره آنلاین | 137 |                              |
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| <a href="https://podspace.pod.ir/api/files/AMSGRN6E5ON7DW74">https://podspace.pod.ir/api/files/AMSGRN6E5ON7DW74</a>                                                                         | Paul Deck                     | Spring MVC: A Tutorial (Second Edition)                                                | کتاب        | 145 |                                    |
| <a href="https://www.coursera.org/learn/software-architecture">https://www.coursera.org/learn/software-architecture</a>                                                                     | Coursera                      | Software Architecture                                                                  | دوره آنلاین | 146 |                                    |
| <a href="https://www.udemy.com/course/master-microservices-with-spring-docker-kubernetes/">https://www.udemy.com/course/master-microservices-with-spring-docker-kubernetes/</a>             | Udemy                         | Master Microservices with Spring Docker Kubernetes                                     | دوره آنلاین | 147 |                                    |
| <a href="https://www.udemy.com/course/event-driven-microservices-with-cqrs-saga-event-sourcing/">https://www.udemy.com/course/event-driven-microservices-with-cqrs-saga-event-sourcing/</a> | Udemy                         | Event Driven Microservices with CQRS Saga Event Sourcing                               | دوره آنلاین | 148 |                                    |
| <a href="https://www.educative.io/courses/introduction-microservice-principles-concepts">https://www.educative.io/courses/introduction-microservice-principles-concepts</a>                 | Educative                     | Introduction to Microservice Principles Concepts                                       | دوره آنلاین | 149 |                                    |
| <a href="https://www.educative.io/courses/domain-driven-design">https://www.educative.io/courses/domain-driven-design</a>                                                                   | Educative                     | Domain Driven Design                                                                   | دوره آنلاین | 150 |                                    |
| <a href="https://podspace.pod.ir/api/files/10H6BBUJM7PA7WI1">https://podspace.pod.ir/api/files/10H6BBUJM7PA7WI1</a>                                                                         | Steve McConnell               | Code Complete: A Practical Handbook of Software Construction                           | کتاب        | 151 | Software Development Methodologies |
| <a href="https://podspace.pod.ir/api/files/CC5BMYLKPTFEM97C">https://podspace.pod.ir/api/files/CC5BMYLKPTFEM97C</a>                                                                         | Robert C. Martin              | Clean Code: A Handbook of Agile Software Craftsmanship                                 | کتاب        | 14  |                                    |
| <a href="https://podspace.pod.ir/api/files/AQXYURY4MWAU79DC">https://podspace.pod.ir/api/files/AQXYURY4MWAU79DC</a>                                                                         | Andrew Hunt and David Thomas  | The Pragmatic Programmer: Your Journey to Mastery                                      | کتاب        | 152 |                                    |
| <a href="https://podspace.pod.ir/api/files/BEUUOYJ2MMLFI27S">https://podspace.pod.ir/api/files/BEUUOYJ2MMLFI27S</a>                                                                         | Jeff Sutherland               | Scrum :The Art of Doing Twice the Work in Half the Time                                | کتاب        | 153 |                                    |
| <a href="https://www.coursera.org/learn/software-processes">https://www.coursera.org/learn/software-processes</a>                                                                           | Coursera                      | Software Processes                                                                     | دوره آنلاین | 154 |                                    |
| <a href="https://www.coursera.org/learn/agile-development-and-scrum">https://www.coursera.org/learn/agile-development-and-scrum</a>                                                         | Coursera                      | Agile Development and Scrum                                                            | دوره آنلاین | 155 |                                    |
| <a href="https://www.scrum.org/resources/blog/30-books-scrum-master">https://www.scrum.org/resources/blog/30-books-scrum-master</a>                                                         | Coursera                      | ۳۰ Books Scrum Master                                                                  | دوره آنلاین | 156 |                                    |
| <a href="https://podspace.pod.ir/api/files/DYGM3C4ZM1UI6LER">https://podspace.pod.ir/api/files/DYGM3C4ZM1UI6LER</a>                                                                         | Nickoloff and Stephen Kuenzli | Docker in Action                                                                       | کتاب        | 157 |                                    |

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| <a href="https://podspace.pod.ir/api/files/JB1EJEPOKVAIWR33">https://podspace.pod.ir/api/files/JB1EJEPOKVAIWR33</a>                                                                                                                         | Nigel Poulton                      | Docker Deep Dive                                 | کتاب        | 158 | Container |
| <a href="https://podspace.pod.ir/api/files/JGF6R297JJXRR2RZ">https://podspace.pod.ir/api/files/JGF6R297JJXRR2RZ</a>                                                                                                                         | Nigel Poulton and Pushkar Joglekar | The Kubernetes Book                              | کتاب        | 159 |           |
| <a href="https://www.simonandschuster.com/books/Kubernetes-in-Action-Second-Edition/Marko-Luksa/In-Action/9781617297618">https://www.simonandschuster.com/books/Kubernetes-in-Action-Second-Edition/Marko-Luksa/In-Action/9781617297618</a> | Marko Luksa                        | Kubernetes in Action                             | کتاب        | 160 |           |
| <a href="https://www.udemy.com/course/docker-mastery/">https://www.udemy.com/course/docker-mastery/</a>                                                                                                                                     | Udemy                              | Docker Mastery                                   | دوره آنلاین | 161 |           |
| <a href="https://www.coursera.org/learn/ibm-containers-docker-kubernetes-openshift">https://www.coursera.org/learn/ibm-containers-docker-kubernetes-openshift</a>                                                                           | Coursera                           | IBM Containers Docker Kubernetes Openshift       | دوره آنلاین | 162 |           |
| <a href="https://www.docker.com/101-tutorial/">https://www.docker.com/101-tutorial/</a>                                                                                                                                                     | Docker                             | Docker 1۰1 Tutorial                              | دوره آنلاین | 163 |           |
| <a href="https://kubernetes.io/training/">https://kubernetes.io/training/</a>                                                                                                                                                               | Kubernetes                         | Kubernetes Training                              | دوره آنلاین | 164 |           |
| <a href="https://podspace.pod.ir/api/files/JG8YWW2IHBOHA44J">https://podspace.pod.ir/api/files/JG8YWW2IHBOHA44J</a>                                                                                                                         | Josh Long and Kenny Bastani        | Cloud Native Java                                | کتاب        | 165 | Cloud     |
| <a href="https://podspace.pod.ir/api/files/ICML46B1S1ZXOTDM">https://podspace.pod.ir/api/files/ICML46B1S1ZXOTDM</a>                                                                                                                         | Action by John Carnell             | Spring Microservices in Action                   | کتاب        | 166 |           |
| <a href="https://podspace.pod.ir/api/files/FQEJSADSOVG97JZQ">https://podspace.pod.ir/api/files/FQEJSADSOVG97JZQ</a>                                                                                                                         | Martin Stefanko                    | Quarkus in Action                                | کتاب        | 167 |           |
| <a href="https://podspace.pod.ir/api/files/H9XG1V4QZL2F71QV/">https://podspace.pod.ir/api/files/H9XG1V4QZL2F71QV/</a>                                                                                                                       | Chris Kemp and Brad Gyger          | Professional Heroku Programming                  | کتاب        | 168 |           |
| <a href="https://podspace.pod.ir/api/files/5U2QFNEKWAGJMV8C">https://podspace.pod.ir/api/files/5U2QFNEKWAGJMV8C</a>                                                                                                                         | Duncan C. E. Winn                  | Cloud Foundry: The Definitive Guide              | کتاب        | 169 |           |
| <a href="https://www.oreilly.com/library/view/apache-zookeeper-essentials/9781784391324/">https://www.oreilly.com/library/view/apache-zookeeper-essentials/9781784391324/</a>                                                               | Saurav Haloi                       | Apache ZooKeeper Essentials                      | کتاب        | 170 |           |
| <a href="https://www.coursera.org/learn/google-cloud-java-spring">https://www.coursera.org/learn/google-cloud-java-spring</a>                                                                                                               | Coursera                           | Google Cloud Java Spring                         | دوره آنلاین | 171 |           |
| <a href="https://www.udemy.com/topic/spring-cloud/">https://www.udemy.com/topic/spring-cloud/</a>                                                                                                                                           | Udemy                              | Spring Cloud                                     | دوره آنلاین | 172 |           |
| <a href="https://www.baeldung.com/spring-cloud-series">https://www.baeldung.com/spring-cloud-series</a>                                                                                                                                     | Baeldung                           | Spring Cloud Series                              | دوره آنلاین | 173 |           |
| <a href="https://www.udemy.com/course/heroku-masterclass/">https://www.udemy.com/course/heroku-masterclass/</a>                                                                                                                             | Udemy                              | Heroku Masterclass                               | دوره آنلاین | 174 |           |
| <a href="https://www.cloudfoundry.org/get-started/">https://www.cloudfoundry.org/get-started/</a>                                                                                                                                           | Cloud Foundry                      | Get Started with Cloud Foundry                   | دوره آنلاین | 175 |           |
| <a href="https://www.baeldung.com/java-dropwizard">https://www.baeldung.com/java-dropwizard</a>                                                                                                                                             | Baeldung                           | Java Dropwizard                                  | دوره آنلاین | 176 |           |
| <a href="https://zookeeper.apache.org/">https://zookeeper.apache.org/</a>                                                                                                                                                                   | ZooKeeper                          | ZooKeeper                                        | دوره آنلاین | 177 |           |
| <a href="https://www.udemy.com/course/apache-zookeeper-tutorial-from-scratch/">https://www.udemy.com/course/apache-zookeeper-tutorial-from-scratch/</a>                                                                                     | Udemy                              | Apache ZooKeeper Tutorial from Scratch           | دوره آنلاین | 178 |           |
| <a href="https://podspace.pod.ir/api/files/LJ16FQYBSR1Q84OG">https://podspace.pod.ir/api/files/LJ16FQYBSR1Q84OG</a>                                                                                                                         | Sander Rossel                      | Continuous Integration, Delivery, and Deployment | کتاب        | 180 |           |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------|-----|-----------|
| <a href="https://podspace.pod.ir/api/files/18LJ76I6UXJPGPSP">https://podspace.pod.ir/api/files/18LJ76I6UXJPGPSP</a>                                                                                                               | Jez Humble and David Farley                                 | Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation | کتاب        | 181 | CI/CD     |
| <a href="https://podspace.pod.ir/api/files/FU78RVFKYUAJGSN7">https://podspace.pod.ir/api/files/FU78RVFKYUAJGSN7</a>                                                                                                               | Nikhil Pathania                                             | Learning Continuous Integration with Jenkins                                                   | کتاب        | 182 |           |
| <a href="https://www.coursera.org/learn/continuous-integration-and-continuous-delivery-ci-cd">https://www.coursera.org/learn/continuous-integration-and-continuous-delivery-ci-cd</a>                                             | Coursera                                                    | Continuous Integration And Continuous Delivery                                                 | دوره آنلاین | 183 |           |
| <a href="https://thectoclub.com/news/best-ci-cd-courses/">https://thectoclub.com/news/best-ci-cd-courses/</a>                                                                                                                     | CTO Club                                                    | CI/CD Course                                                                                   | دوره آنلاین | 184 |           |
| <a href="https://www.jenkins.io/doc/tutorials/">https://www.jenkins.io/doc/tutorials/</a>                                                                                                                                         | Jenkins                                                     | Jenkins Tutorial                                                                               | دوره آنلاین | 185 |           |
| <a href="https://www.coursera.org/specializations/jenkins-zero-to-hero">https://www.coursera.org/specializations/jenkins-zero-to-hero</a>                                                                                         | Coursera                                                    | Jenkins Zero to Hero                                                                           | دوره آنلاین | 186 |           |
| <a href="https://discuss.circleci.com/t/introducing-circleci-academy-our-developer-focused-learning-platform/34709">https://discuss.circleci.com/t/introducing-circleci-academy-our-developer-focused-learning-platform/34709</a> | Circle CI                                                   | Introducing Circle CI Academy                                                                  | دوره آنلاین | 187 |           |
| <a href="https://www.udemy.com/course/argo-cd-lightning-course/">https://www.udemy.com/course/argo-cd-lightning-course/</a>                                                                                                       | Udemy                                                       | Argo Lightning Course                                                                          | دوره آنلاین | 188 |           |
| <a href="https://podspace.pod.ir/api/files/1DQ15UUQBZ3U2">https://podspace.pod.ir/api/files/1DQ15UUQBZ3U2</a>                                                                                                                     | Bruce Snyder, Dejan Bosanac and Rob Davies                  | ActiveMQ in Action                                                                             | کتاب        | 189 | Messaging |
| <a href="https://podspace.pod.ir/api/files/BDW78CCH6M5JQM9K">https://podspace.pod.ir/api/files/BDW78CCH6M5JQM9K</a>                                                                                                               | Dylan Scott, Viktor Gamov and Dave Klein                    | Kafka in Action                                                                                | کتاب        | 190 |           |
| <a href="https://podspace.pod.ir/api/files/F3I9PSGCAS3XWXC�">https://podspace.pod.ir/api/files/F3I9PSGCAS3XWXC�</a>                                                                                                               | Mark Richards, Richard Monson-Haefel, and David A. Chappell | Java Message Service                                                                           | کتاب        | 191 |           |
| <a href="https://podspace.pod.ir/api/files/36HQ9KUXQDVDJPJV">https://podspace.pod.ir/api/files/36HQ9KUXQDVDJPJV</a>                                                                                                               | Neha Narkhede, Gwen Shapira, and Todd Palino                | Kafka :The Definitive Guide                                                                    | کتاب        | 192 |           |
| <a href="https://www.udemy.com/course/java-message-service-jms-fundamentals/">https://www.udemy.com/course/java-message-service-jms-fundamentals/</a>                                                                             | Udemy                                                       | Java Message Service JMS Fundamentals                                                          | دوره آنلاین | 193 |           |
| <a href="https://www.tutorialspoint.com/apache_activemq/index.htm">https://www.tutorialspoint.com/apache_activemq/index.htm</a>                                                                                                   | TutorialsPoint                                              | Apache Activemq                                                                                | دوره آنلاین | 194 |           |
| <a href="https://activemq.apache.org/components/artemis/documentation/">https://activemq.apache.org/components/artemis/documentation/</a>                                                                                         | ActiveMq                                                    | Artemis                                                                                        | دوره آنلاین | 195 |           |
| <a href="https://www.udemy.com/course/kafka-fundamentals-for-java-developers/">https://www.udemy.com/course/kafka-fundamentals-for-java-developers/</a>                                                                           | Udemy                                                       | Kafka Fundamentals for Java Developers                                                         | دوره آنلاین | 196 |           |
| <a href="https://www.confluent.io/blog/best-kafka-tutorials-examples-and-learning-resources/">https://www.confluent.io/blog/best-kafka-tutorials-examples-and-learning-resources/</a>                                             | Confluent                                                   | Best kafka Tutorials, Examples and Learning Sources                                            | دوره آنلاین | 197 |           |
| <a href="https://podspace.pod.ir/api/files/2WVDV3O2322INZ14">https://podspace.pod.ir/api/files/2WVDV3O2322INZ14</a>                                                                                                               | Vahe Aslanyan                                               | Java Object Oriented Programming                                                               | کتاب        | 198 |           |



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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------|-------------|-----|-----------------|
| <a href="https://podspace.pod.ir/api/files/K9BBZTJZKD4V7B6Q">https://podspace.pod.ir/api/files/K9BBZTJZKD4V7B6Q</a>                                                                                                               | Robert Lafore                                                                       | Data Structures and Algorithms in Java              | کتاب        | 01  | Data Structures |
| <a href="https://podspace.pod.ir/api/files/1HUOEWPY9DEVQRUY">https://podspace.pod.ir/api/files/1HUOEWPY9DEVQRUY</a>                                                                                                               | Robert Sedgewick and Kevin Wayne                                                    | Algorithms ,Part I and II                           | کتاب        | 02  |                 |
| <a href="https://podspace.pod.ir/api/files/BBJG9P8YWR5E7OXN/">https://podspace.pod.ir/api/files/BBJG9P8YWR5E7OXN/</a>                                                                                                             | Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein (CLRS) | Introduction to Algorithms                          | کتاب        | 03  |                 |
| <a href="https://podspace.pod.ir/api/files/H2F6DOREKCEXV872">https://podspace.pod.ir/api/files/H2F6DOREKCEXV872</a>                                                                                                               | دکتر محمد قدسی                                                                      | داده ساختارها و الگوریتمها                          | کتاب        | 04  |                 |
| <a href="https://www.coursera.org/learn/data-structures-optimizing-performance?specialization=java-object-oriented">https://www.coursera.org/learn/data-structures-optimizing-performance?specialization=java-object-oriented</a> | Coursera                                                                            | Data Structures Optimizing Performance              | دوره آنلاین | 05  |                 |
| <a href="https://www.udemy.com/course/data-structures-and-algorithms-deep-dive-using-java/">https://www.udemy.com/course/data-structures-and-algorithms-deep-dive-using-java/</a>                                                 | Udemy                                                                               | Data Structures and Algorithms Depp Dive Using Java | دوره آنلاین | 199 |                 |
| <a href="https://www.geeksforgeeks.org/data-structures/">https://www.geeksforgeeks.org/data-structures/</a>                                                                                                                       | GeeksForGeeks                                                                       | Data Structures                                     | دوره آنلاین | 07  |                 |
| <a href="https://www.mygreatlearning.com/academy/learn-for-free/courses/data-structures-and-algorithms-in-java">https://www.mygreatlearning.com/academy/learn-for-free/courses/data-structures-and-algorithms-in-java</a>         | My Great Learning                                                                   | Data Structures and Algorithms in Java              | دوره آنلاین | 200 |                 |



## ۲ بخش دوم: مسیرهای یادگیری تفکیکی

## ۲/۱ مسیر یادگیری سطح کاردان

هدف اصلی این سطح: یادگیری اصول بنیادی، کدنویسی تمیز، و آشنایی با ابزارها و ساختار پروژه‌های شرکت.

| جدول ۲: مسیر یادگیری سطح کاردان                              |                               |                                              |                 |                            |
|--------------------------------------------------------------|-------------------------------|----------------------------------------------|-----------------|----------------------------|
| هدف یادگیری                                                  | فصل‌ها / بخش‌های مورد نیاز    | عنوان منبع                                   | شناسه منبع (ID) | حوزه                       |
| types & syntax Basic                                         | Ch 1-2-3                      | Head First Java                              | 18              | Core Java                  |
| pillars (class / interface / OOP inheritance / polymorphism) | Ch 3-4                        | Head First Java                              | 18              | Core Java                  |
| modules & Packages                                           | Ch 4.8                        | Core Java                                    | 19              | Core Java                  |
| Exceptions                                                   | Ch 11                         | Head First Java                              | 18              | Core Java                  |
| (List, Map, Set) Collections                                 | Ch 16                         | Head First Java                              | 18              | Core Java                  |
| (basics) Generics                                            | Ch 16                         | Head First Java                              | 18              | Core Java                  |
| (intro) Lambdas                                              | Ch 1                          | Modern Java in Action                        | 29              | Core Java                  |
| terminal ops (forEach, filter, – Streams map)                | Ch 2                          | Modern Java in Action                        | 29              | Core Java                  |
| basics (SLF4J) Logging                                       | Ch 1                          | Java Logging                                 | 120             | Core Java                  |
| POM basics Maven                                             | Ch 3                          | Maven :The Definitive Guide                  |                 | Core Java                  |
| basics (@Test, assertions) JUnit 5                           | Sec 2-3                       | JUnit 5 User Guide                           |                 | Test Automation            |
| intuition ( $O(1)$ , $O(\log n)$ , $O(n)$ ) Big-O            | notes ۱ MJ Week               | Algorithms (Princeton Coursera) Part I       | 02              | DS & Algorithms            |
| vs Lists Arrays                                              | Ch 2                          | Data Structures & Algorithm Analysis in Java |                 | DS & Algorithms            |
| HashSet internals / HashMap                                  | Ch 3.4                        | Algorithms (Sedgewick & Wayne)               | 02              | DS & Algorithms            |
| APIs java.util Searching with & Sorting                      | (sorting) Ch 2                | Algorithms (Sedgewick & Wayne)               | 02              | DS & Algorithms            |
| Boot scaffold Spring (SpringBootApplication@)                | Ch 1                          | Spring in Action                             | 52              | Spring                     |
| CRUD endpoints + RestController@                             | Ch 2                          | Spring in Action                             | 52              | Spring                     |
| Autowired@ / DI Constructor                                  | Ch 2                          | Spring in Action                             | 52              | Spring                     |
| CrudRepository Data JPA Spring                               | Ch 3                          | Spring in Action                             | 52              | Spring                     |
| SELECT / INSERT Basic                                        | Ch 1-4                        | Learning SQL                                 | 74              | Database                   |
| JOIN Simple                                                  | Ch 5                          | Learning SQL                                 | 74              | Database                   |
| commit / push / Clone                                        | Ch 2                          | Pro Git                                      | 67              | Version Control            |
| PR flow & Branch                                             | Ch 3                          | Pro Git                                      | 67              | Version Control            |
| simple conflict Resolve                                      | Basic Merge “ Ch 3 “Conflicts | Pro Git                                      | 67              | Version Control            |
| build + test Maven                                           | Ch 3                          | Maven :The Definitive Guide                  |                 | CI & Build                 |
| pipeline (GitHub/GitLab Simple Actions)                      | with Maven” Java” starter     | GitHub Actions Docs                          |                 | CI & Build                 |
| board basics (story, task, bug) Agile                        | Guide Scrum (p 16) 2020       | Scrum Guide                                  | 153             | Software Dev Methodologies |

|                       |                                  |                              |  |                               |
|-----------------------|----------------------------------|------------------------------|--|-------------------------------|
| review etiquette Code | Review Code“<br>”Developer Guide | Engineering Practices Google |  | Software Dev<br>Methodologies |
|-----------------------|----------------------------------|------------------------------|--|-------------------------------|

## ۲/۲ مسیر یادگیری سطح کارشناس

هدف اصلی این سطح: عمیق‌تر شدن در تکنولوژی‌ها، کسب استقلال در انجام تسک‌ها، و شروع به درک مفاهیم معماری.

| جدول ۳: مسیر یادگیری سطح کارشناس |                 |                                 |                            |                                                          |
|----------------------------------|-----------------|---------------------------------|----------------------------|----------------------------------------------------------|
| حوزه                             | شناسه منبع (ID) | عنوان منبع                      | فصل‌ها / بخش‌های مورد نیاز | هدف یادگیری                                              |
| Data Structures                  | 19              | Core Java                       | Ch 3.10                    | Arrays                                                   |
| Data Structures                  | 19              | Core Java                       | Ch 9                       | Lists (ArrayList / LinkedList)                           |
| Data Structures                  | 19              | Core Java                       | Ch 9                       | Sets (HashSet vs TreeSet)                                |
| Data Structures                  | 19              | Core Java                       | Ch 9                       | Maps (HashMap vs TreeMap)                                |
| Algorithms                       | 18              | Head First Java                 | Ch 3-4                     | Sorting with Collections.sort / Arrays.sort              |
| Algorithms                       | 19              | Core Java                       | Ch 9 & 6                   | Binary Search & Collections.binarySearch                 |
| Algorithms                       |                 | The Algorithm Design Manual     | Ch 2                       | Big-O refresher ( $O(1)$ , $O(\log n)$ , $O(n \log n)$ ) |
| Object Oriented                  |                 | Head First OOAD                 | Ch 6-7                     | Abstraction & Encapsulation                              |
| Object Oriented                  |                 | Head First OOAD                 | Ch 6-7                     | Inheritance vs Composition                               |
| Object Oriented                  |                 | Head First OOAD                 | Ch 6-7                     | Polymorphism in Java                                     |
| Clean Code                       |                 | Clean Code                      | Ch 2-3                     | Recognising Code Smells                                  |
| Clean Code                       |                 | Refactoring                     | Ch 1-2                     | Small Refactorings (Rename, Extract Method)              |
| Java Core                        | 29              | Modern Java in Action           | Ch 40-                     | Streams collectors (toList, groupingBy)                  |
| Java Core                        | 29              | Modern Java in Action           | Ch 3                       | Lambdas in practice                                      |
| Java Core                        | 29              | Modern Java in Action           | Ch 11                      | Optionals for null-safety                                |
| Java Core                        | 29              | Modern Java in Action           | Ch 1-2                     | Functional Interfaces                                    |
| Java Core                        |                 | Java: A Beginners Guide         | Ch 13                      | Generics (wildcards, bounds)                             |
| Java Core                        |                 | Java: A Beginners Guide         | Ch 9                       | Exception Hierarchy & custom exceptions                  |
| Java Core                        | 19              | Core Java                       | Ch 9                       | Collections API tuning (initialCapacity, loadFactor)     |
| Java Advanced                    | 28              | Java Concurrency in Practice    | Ch 6                       | ExecutorService basics (newFixedThreadPool)              |
| Java Advanced                    | 19              | Core Java                       | Ch 12                      | CompletableFuture intro                                  |
| Design Patterns                  |                 | Head First Design Patterns      | Ch 4                       | Factory Method                                           |
| Design Patterns                  |                 | Head First Design Patterns      | Ch 5                       | Singleton (lazy vs eager)                                |
| Design Patterns                  |                 | Head First Design Patterns      | Ch 1                       | Strategy                                                 |
| ORM                              | 41              | Java Persistence with Hibernate | Ch 4                       | Entity Lifecycle (new, managed, detached, removed)       |
| ORM                              | 41              | Java Persistence with Hibernate | Ch 5                       | JPQL basics (SELECT e FROM...)                           |
| ORM                              | 41              | Java Persistence with Hibernate | Ch 6                       | Pagination with setFirstResult / setMaxResults           |
| ORM                              | 41              | Java Persistence with Hibernate | Ch 4                       | 1st-level cache behaviour                                |

|                                               |                         |                                       |     |                  |
|-----------------------------------------------|-------------------------|---------------------------------------|-----|------------------|
| Boot Autoconfiguration basics                 | Ch 3                    | Spring in Action                      | 52  | Spring           |
| Controllers & Validation (@Valid)             | Ch 4                    | Spring in Action                      | 52  | Spring           |
| Bean scopes (singleton ,prototype)            | Ch 4                    | Spring in Action                      | 52  | Spring           |
| Spring Data JPA (CrudRepository)              | Ch 5                    | Spring in Action                      | 52  | Spring           |
| RESTful resource design                       | Ch 4                    | RESTful Web Services                  | 59  | Web Service      |
| API versioning (/v1.../)                      | Ch 5                    | RESTful Web Services                  | 59  | Web Service      |
| OpenAPI / Swagger basics                      | Overview                | OpenAPI Specification docs            |     | Web Service      |
| Interactive Rebase (git rebase -i)            | Ch 3                    | Pro Git                               | 67  | Version Control  |
| Merge strategies (--squash-- ,no-ff)          | Ch 3                    | Pro Git                               | 67  | Version Control  |
| INNER / LEFT JOIN review                      | Ch 3-4                  | Learning SQL                          | 74  | Database         |
| Index basics & EXPLAIN                        | Ch 8                    | Learning SQL                          | 74  | Database         |
| Maven lifecycle (compile ,test ,package)      | Ch 4                    | Maven :The Definitive Guide           |     | Build & CI       |
| Basic GitHub Actions pipeline (mvn test)      | "Java with Maven" guide | GitHub Docs — "Java with Maven" guide |     | Build & CI       |
| Dockerfile for Spring Boot JAR                | Ch 2                    | Docker in Practice                    | 157 | Build & CI       |
| Spring Boot Test (@SpringBootTest)            | Part II                 | Test-Driven Java Development          | 93  | Test Automation  |
| Mockito stubbing & spies                      | Ch 2                    | Mockito Cookbook                      |     | Test Automation  |
| Spring Security HTTP Basic                    | Ch 2                    | Spring Security in Action             | 107 | Auth & AuthZ     |
| Form login with UserDetailsService            | Ch 3                    | Spring Security in Action             | 107 | Auth & AuthZ     |
| Actuator metrics endpoint                     | "Actuator"              | Spring Boot Reference                 |     | Monitoring       |
| Log aggregation basics (ELK)                  | Ch 3                    | The Art of Monitoring                 |     | Monitoring       |
| Simple in-memory cache with ConcurrentHashMap | Ch ۵/۲-۵/۱۶             | Java Concurrency in Practice          | 28  | Caching          |
| Continuous Delivery pipeline concepts         | Ch 6                    | The DevOps Handbook                   |     | SD Methodologies |
| Writing a Dockerfile (FROM openjdk...)        | Ch 2                    | Docker in Practice                    | 157 | Container        |
| Docker Compose for dev (docker-compose up)    | Ch 5                    | Docker in Practice                    | 157 | Container        |
| Publish/Subscribe with ActiveMQ               | Ch 3                    | ActiveMQ in Action                    | 189 | Messaging        |

## ۲/۳ مسیر یادگیری سطح کارشناس ارشد

هدف اصلی این سطح: تسلط بر معماری سیستم‌های توزیع‌شده، توانایی منتورینگ دیگران، و راهبری فنی پروژه‌ها.

| جدول ۴: مسیر یادگیری سطح کارشناس ارشد |                            |                                                              |                 |                 |
|---------------------------------------|----------------------------|--------------------------------------------------------------|-----------------|-----------------|
| هدف یادگیری                           | فصل‌ها / بخش‌های مورد نیاز | عنوان منبع                                                   | شناسه منبع (ID) | حوزه            |
| Dynamic Programming (Knapsack)        | Ch 15 / Ch 6               | Intro to Algorithms / Algorithm Design Manual                | -- / --         | Algorithms      |
| Tactical & Strategic DDD              | Ch 1-2 / Ch 3-4            | Domain-Driven Design / Implementing DDD                      | 142 / 143       | Object Oriented |
| Clean Architecture at scale           | Ch 10 & 14 / Ch 7          | Clean Code / Clean Architecture                              | -- / 141        | Clean Code      |
| Custom Annotations                    | Ch 24-25 / Item 39         | Java :The Complete Reference / Effective Java                | 26 / 13         | Java Core       |
| Reflection deep-dive & perf tips      | Ch 26 / Item 65            | Java :The Complete Reference/ Effective Java                 | 26 / 13         | Java Core       |
| Records                               | Ch 1 "Records"             | Java 17 Cookbook                                             |                 | Java Core       |
| Switch Expressions                    | Ch 15                      | Modern Java in Action                                        | 29              | Java Core       |
| Pattern Matching (instanceof, switch) | "Pattern Matching"         | Inside Java blog series                                      |                 | Java Core       |
| Text Blocks("""" ... """)             | Ch 15                      | Modern Java in Action                                        | 29              | Java Core       |
| var & implicit typing best-practice   | Item 64                    | Effective Java                                               | 13              | Java Core       |
| Fork/Join & parallel streams          | Ch 7 / Item 32             | Java Concurrency in Practice / Effective Java                | 28 / 13         | Java Advanced   |
| CompletableFuture chaining & errors   | Ch 7 / Ch 11               | Java Concurrency in Practice / Modern Java in Action         | 28 / 29         | Java Advanced   |
| Java Memory Model & volatile          | Ch 16                      | Java Concurrency in Practice                                 | 28              | Java Advanced   |
| Enterprise Integration patterns       | Ch 12 / Ch 4               | Head First Design Patterns / Enterprise Integration Patterns |                 | Design Patterns |
| CQRS pattern                          | Ch 7 / Ch 10               | PoEAA / Implementing DDD                                     | 36 / 143        | Design Patterns |
| Entity lifecycle (states)             | Ch 4                       | Java Persistence with Hibernate                              | 41              | ORM / JPA       |
| JPQL pagination (setMaxResults)       | Ch 6                       | Java Persistence with Hibernate                              | 41              | ORM / JPA       |
| N+1 query detection & fix             | Ch 3                       | High-Performance Hibernate                                   |                 | ORM / JPA       |
| Batch inserts / updates               | Ch 4                       | High-Performance Hibernate                                   |                 | ORM / JPA       |
| Config Server                         | Ch 12                      | Spring in Action                                             | 52              | Spring Cloud    |

|                                          |                                                                   |                                    |         |                     |
|------------------------------------------|-------------------------------------------------------------------|------------------------------------|---------|---------------------|
| REST versioning strategies               | Ch 5                                                              | RESTful Web Services               | 59      | Web Service / REST  |
| OpenAPI / Swagger design                 | Overview /Ch 5                                                    | OpenAPI Spec /RESTful Web Services | -- / 59 | Web Service / REST  |
| Rebase workflow (rebase -onto)           | Ch 3                                                              | Pro Git                            | 67      | Version Control     |
| Advanced merge flags (--squash-- ,no-ff) | Ch 3                                                              | Pro Git                            | 67      | Version Control     |
| Index design & EXPLAIN plans             | Ch 8                                                              | Learning SQL                       | 74      | Database            |
| Transaction isolation levels             | Ch 20                                                             | SQL Antipatterns                   |         | Database            |
| Query optimisation workflow              | Ch 3                                                              | SQL Performance Explained          |         | Database            |
| Maven multi-module & profiles            | Ch 5-6                                                            | Maven :The Complete Reference      |         | Build Tools         |
| Gradle multi-project build               | Ch 4                                                              | Gradle in Action                   | 85      | Build Tools         |
| @SpringBootTest integration tests        | Part III                                                          | TDD Java Development               |         | Test Automation     |
| Testcontainers for DB integration        | "Getting Started"                                                 | Testcontainers docs                |         | Test Automation     |
| JUnit 5 parameterised / dynamic tests    | Section 0                                                         | JUnit 5 User Guide                 |         | Test Automation     |
| OAuth 2 Authorization Code flow          | Ch 9                                                              | Spring Security in Action          | 107     | Auth & AuthZ        |
| JWT claims & validation                  | Ch 10                                                             | Spring Security in Action          | 107     | Auth & AuthZ        |
| Spring Boot Actuator metrics             | "Actuator"                                                        | Spring Boot Reference              |         | Monitoring          |
| Log aggregation with ELK                 | Ch 3                                                              | Art of Monitoring                  |         | Monitoring          |
| Redis cache integration                  | Ch 8                                                              | Redis in Action                    | 131     | Caching             |
| Continuous Delivery patterns             | Ch 9                                                              | DevOps Handbook                    |         | SD Methodologies    |
| Docker Compose for local microservices   | Ch 5                                                              | Docker in Practice                 | 157     | Container / Dev Env |
| Load-Balancer / Reverse-Proxy basics     | Ch 6                                                              | Building Microservices             |         | System Design       |
| Kafka exactly-once semantics             | Ch 4                                                              | Kafka :The Definitive Guide        | 192     | Messaging           |
| Retry / dead-letter patterns             | Ch 4                                                              | Kafka :The Definitive Guide        | 192     | Messaging           |
| Maven Goals & Phases                     | Ch 4 (Build Lifecycles)                                           | Maven :The Definitive Guide        |         | Build Tools         |
| Gradle Tasks & Profiles                  | Ch 4 (Multi-project & task graph)                                 | Gradle in Action                   | 85      | Build Tools         |
| Apache Mina basics                       | Ch ۱ ("Getting Started") & Ch ۲ ("Core APIs")                     | Learning Apache MINA) O'Reilly(    |         | Socket Programming  |
| Netty fundamentals                       | Ch ۱ ("Fundamentals") & Ch ۲ ("Bootstrapping a Server")           | Netty in Action                    |         | Socket Programming  |
| OWASP Top ۱۰ awareness                   | "Top ۱۰ Web Application Security Risks"                           | OWASP Foundation                   |         | Security            |
| Cryptography introduction                | Ch ۱ ("Principles of Modern Crypto") & Ch ۲ ("Symmetric Ciphers") | Cryptography Engineering           |         | Security            |



## ۲/۴ مسیر یادگیری سطح عالی

هدف اصلی این سطح: راهبری فنی در سطح شرکت، تعریف استانداردها، تحقیق و توسعه، و آموزش.

| جدول ۵: مسیر یادگیری سطح عالی                            |                            |                                    |                 |                               |
|----------------------------------------------------------|----------------------------|------------------------------------|-----------------|-------------------------------|
| هدف یادگیری                                              | فصل ها / بخش های مورد نیاز | عنوان منبع                         | شناسه منبع (ID) | حوزه                          |
| Advanced DDD patterns (Bounded Context, Event Sourcing)  | Ch 14-15                   | Domain-Driven Design               | 142             | Object Oriented               |
| DDD                                                      | Ch 11-13                   | Implementing DDD                   | 143             | System Design & Architectures |
| Internet-scale architecture principles                   | Ch 10-11                   | Clean Architecture                 | 141             | Clean Code / Architecture     |
| Microservices                                            | Ch 11                      | Building Microservices             |                 | System Design & Architectures |
| GC tuning flags (G1, ZGC)                                | Ch 8                       | Java Performance                   | 27              | Java Advanced                 |
| java-profiler                                            | Ch 7                       | Java Performance                   | 27              | Java Advanced                 |
|                                                          | JFR Essentials e-book      | JFR Essentials e-book              |                 | Java Advanced                 |
| Cloud-native patterns (Circuit Breaker, Saga)            | Ch 6                       | Cloud Native Patterns              | 165             | Design Patterns               |
|                                                          | Ch 8-9                     | Enterprise Integration Patterns    |                 |                               |
| Enterprise Integration (Router, Aggregator)              | Ch 4                       | Enterprise Integration Patterns    |                 | Design Patterns               |
| Reactive Hibernate (Mutiny)                              | Ch 12                      | Java Persistence with Hibernate    | 41              | ORM / JPA                     |
| Service discovery basics                                 | Ch 12                      | Spring in Action                   | 52              | Spring Cloud                  |
| Spring Cloud Gateway & Resilience4j                      | Ch 2-3                     | Spring Cloud Gateway Guide         |                 | Spring                        |
|                                                          | Ch 6                       | Spring Microservices in Action     | 166             | Spring                        |
| Reactive back-pressure (Flux, Mono)                      | Ch 5                       | Reactive Spring                    |                 | Spring                        |
| gRPC bi-directional streaming                            | Ch 4                       | gRPC: Up & Running                 | 60              | Web Services                  |
| CQRS + Event Store read models                           | Ch 11                      | Designing Data-Intensive Apps      |                 | Database                      |
| Gradle custom plugins                                    | Ch 7                       | Gradle in Action                   | 85              | Build Tools                   |
| Twelve-Factor App refresher                              | Ch 2                       | Cloud Native Java                  | 165             | Cloud                         |
| Serverless fundamentals (Function-as-a-Service concepts) | Ch 4-5                     | Serverless Architectures (Manning) |                 | Cloud                         |
| Resilience patterns (Circuit-Breaker, Bulkhead)          | Ch 6-8                     | Cloud Native Patterns              | 165             | Cloud                         |
| Prometheus metrics                                       | Ch 6                       | Prometheus Up & Running            | 128             | Monitoring & Observability    |
| Distributed tracing (OpenTelemetry ...)                  | Ch 4                       | Distributed Tracing in Practice    |                 | Monitoring                    |
| Redis Cluster & eviction policies                        | Ch 8                       | Redis in Action                    | 131             | Caching                       |
| Kafka exactly-once with Idempotent Producer              | Ch 4                       | Kafka :The Definitive Guide        | 192             | Messaging                     |



|                                                      |                                                                                                                           |                                                                             |     |                              |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----|------------------------------|
| GC log analysis with GC Viewer                       | Ch 8                                                                                                                      | Java Performance                                                            | 27  | Performance & Profiling      |
| Service Mesh basics                                  | Ch 14                                                                                                                     | Kubernetes Up & Running                                                     | 159 | System Design & Architecture |
| Micro-frontends overview                             | Ch 1                                                                                                                      | Building Micro-Frontends                                                    |     | Front-end Awareness          |
| Sharded Counter pattern                              | Ch 9) Counter-Sharding(                                                                                                   | Designing Data-Intensive Apps                                               |     | System Design & Architecture |
| Rate-Limiter design                                  | Ch 10 / Cloudflare Blog "Rate Limiting 101"                                                                               | Web Scalability for Startup Engineers / Cloudflare Blog "Rate Limiting 101" |     | System Design & Architecture |
| Performance Load-Test with JMeter                    | Ch 3 (Thread Groups) / Best Practices                                                                                     | Master Apache JMeter (O'Reilly) / Apache JMeter User Manual                 |     | Test Automation              |
| Netty performance tuning/Apache Mina custom protocol | Ch") ۵ Advanced Pipeline Concepts & ("Ch") ۶ Performance Tuning"/Ch ۴")Protocol Codec API ("&Ch ۵")High-Performance I/O(" | Netty in Action /Learning Apache MINA                                       |     | Socket Programming           |
| OWASP Proactive Controls                             | "Proactive Controls"                                                                                                      | OWASP Foundation                                                            |     | Security                     |
| Applied Cryptography & key management                | Ch") ۳ Public-Key Primitives & ("Ch ۴")Key Management/("                                                                  | Serious Cryptography                                                        |     | Security                     |

